Morse theory and 2D stationary Euler flows

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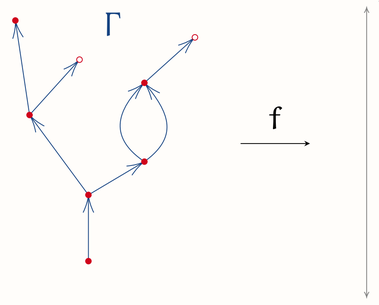
Introduction

The 2D incompressible Euler equations describe the motion of an ideal fluid on a surface M endowed with an area form ω. A flow is called steady if it does not depend on time. Arnold showed in Ref. [1] that the Euler equations can be expressed as geodesic flow on the group of volume-preserving diffeomorphisms, $\mathrm{SDiff}(M)$. Steady flows correspond to critical points of the energy functional on coadjoint orbits. This geometric viewpoint connects hydrodynamics with symplectic geometry and Morse theory, enabling a topological classification of steady Euler flows via Reeb graphs.

Reeb Graphs

A Reeb graph (Γ, f) is an oriented connected graph Γ with a continuous function $f\colon \Gamma \to \mathbb{R}$ such that

- each vertex has degree 1 or 3,
- a degree 1 vertex can be boundary $\circ$ or non boundary $\bullet$,
- for vertices with degree 3 there are either two incoming and one outgoing edge, or vice versa,
- f is strictly monotonous on each edge of Γ, and the edges of Γ are oriented towards the direction of increasing f.



Measured Reeb graphs

A Reeb graph (Γ, f) endowed with a log-smooth measure μ is called a measured Reeb graph (Γ, f, μ), and we say it is compatible with (M,ω) if

- the genus of M is equal to $\dim H_1(\Gamma,\mathbb{R})$,
- the number of boundary vertices of Γ is equal to the number of connected components of ∂M,
- the volume of Γ with respect to the measure μ is equal to the volume of M,

$$\int_\Gamma d\mu = \int_M \omega.$$

Circulation Reeb graphs

We define a density ρ_f in (Γ, f, μ) as $\rho_f(I) \coloneqq \int_I f\, d\mu$. A measured Reeb graph endowed with a circulation function is called a circulation Reeb graph (Γ, f, μ, c), where c is an antiderivative of the density ρ_f.

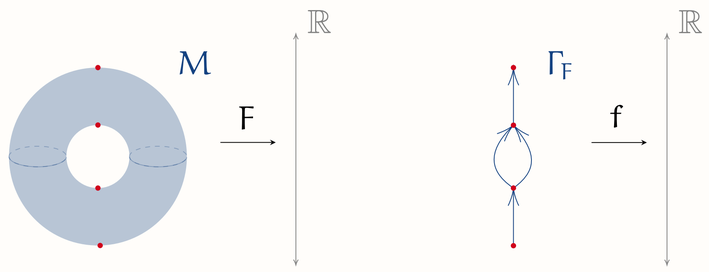
Classification of...

...Morse functions

A simple Morse function $F\colon M \to \mathbb{R}$ on a surface is a differentiable function such that:

- all its critical points are non-degenerate,
- does not have critical points at the boundary,
- is constant in each connected component of the boundary,
- each level contains at most one critical point.

We can associate to each simple Morse function F on M a measured Reeb graph compatible with (M, ω) as follows. The graph Γ_F is defined as the quotient of the surface M by the connected components of the level sets of F. The function f is the one induced by F on the quotient. The natural projection map $\pi\colon M \to \Gamma_F$ induces a measure μ_F on Γ_F, defined as the pushforward of the area form on M under π.



This gives a one-to-one correspondence between simple Morse functions on M, considered up to symplectomorphisms, and measured Reeb graphs compatible with (M,ω) considered up to isomorphisms.

Simple Morse functions on M	$\xrightarrow{\sim}$	Measured Reeb graphs compatible with (M, ω)
$F\colon M \to \mathbb{R}$	$\longmapsto$	(Γ_F, f, μ_F)

...Morse-type cosets of one-forms

A coset of one-forms $[\alpha] \in \mathfrak{svect}^*(M)$ is Morse-type if $\mathfrak{curl}[\alpha] = d\alpha/\omega$ is a simple Morse function. We can associate to each Morse-type coset $[\alpha]$ on M a circulation Reeb graph compatible with (M, ω) as follows. We take the function $F = d\alpha/\omega$ and construct the measured Reeb graph (Γ_F, f, μ_F). We define the circulation function c_F by

$$c_F(x) \coloneqq \int_{\pi^{-1}(x)} \alpha.$$

We denote $\Gamma_{[\alpha]} = (\Gamma_F, f, \mu_F, c_F)$. This gives a one-to-one correspondence between Morse-type cosets of one-forms, considered up to belonging to the same orbit of the $\mathrm{SDiff}(M)$ coadjoint action, and measured Reeb graphs compatible with (M,ω) considered up to isomorphisms.

Morse-type cosets of one forms	$\xrightarrow{\sim}$	Circulation Reeb graphs compatible with (M, ω)
$[\alpha]$	$\longmapsto$	$\Gamma_{[\alpha]}$

...Morse-type coadjoint orbits

An important property of the mapping $\mathfrak{curl}$ is its equivariance with respect to the $\mathrm{SDiff}(M)$ action: if cosets $[\alpha]$ and $[\beta]$ belong to the same coadjoint orbit, then the functions $d\alpha/\omega$ and $d\beta/\omega$ are conjugated by a symplectic diffeomorphism. In particular, if $d\alpha/\omega$ is a simple Morse function, then so is $d\beta/\omega$. So, a coadjoint orbit $\mathcal{O} \subset \mathfrak{svect}^*(M)$ is Morse-type if any coset $[\alpha] \in \mathcal{O}$ is Morse-type.

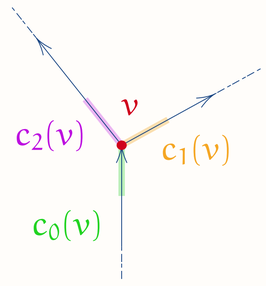
Then, the previous classification gives a one-to-one correspondence between Morse-type coadjoint orbits of $\mathrm{SDiff}(M)$ and circulation Reeb graphs compatible with (M,ω) considered up to isomorphisms

Morse-type coadjoint orbits of $\mathrm{SDiff}(M)$	$\longrightarrow$	Circulation Reeb graphs compatible with (M, ω) up to isomorphism
$\mathcal{O}$	$\longmapsto$	(Γ, f, μ, c)

Steady Flows Characterization

Let v be a degree 3 vertex of a circulation Reeb graph, and e_0, e_1, e_2 the corresponding edges. We denote

$$c_i(v) = \lim_{x \xrightarrow[e_i]{} v} c(x).$$



We say that c is balanced if, for every degree 3 vertex, the numbers $c_0(v)$, $c_1(v)$, and $c_2(v)$ are non-zero and have the same sign. Then we have the next characterization:

Theorem: Let $\mathcal{O} \subset \mathfrak{svect}^*(M)$ be a Morse-type coadjoint orbit. Then $\mathcal{O}$ admits a steady flow if and only if the corresponding circulation function is balanced.

Morse-type coadjoint orbits of $\mathrm{SDiff}(M)$	$\longrightarrow$	Circulation Reeb graphs compatible with (M, ω) up to isomorphism
$\mathcal{O}$	$\longmapsto$	(Γ, f, μ, c)
$\mathcal{O}$ admits a steady flow	$\longleftrightarrow$	c is balanced

Closed surfaces

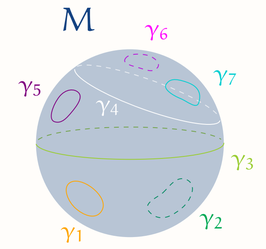
If M is a closed surface, that is, without boundary, the condition of being balanced can be replaced with totally negative, that means that the sign of $c_0(v)$, $c_1(v)$ and $c_2(v)$ is negative.

Theorem: Let (Γ, f, μ) be a measured Reeb graph with no boundary vertices. Then, the set of totally negative circulation functions on Γ is an open convex polytope in the affine space of all circulation functions.

In particular, the set of totally negative circulation functions on Γ is bounded; hence, most Morse-type coadjoint orbits of $\mathrm{SDiff}(M)$ do not admit steady solutions of the Euler equation.

Application

Given any finite collection $\gamma_1, \ldots, \gamma_n$ of pairwise disjoint closed embedded curves in a surface M, does there exist a steady solution to the Euler equations on M having the γ_i's as streamlines?



The theory developed in this poster seems to be a powerful tool to prove this realization theorem.

References

[1] V.I. Arnold, B.A. Khesin, *Topological Methods in Hydrodynamics*, 2nd ed., Springer, 2021.

[2] A. Izosimov, B. Khesin, *Characterization of Steady Solutions to the 2D Euler Equation*, Int. Math. Res. Not., 2017(24), 7459–7503.

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